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th National High School Physics Competition: Final Problems

- Beijing

Theory

1.

In football, a player at passes along ; the ball rolls on grass at speed uniformly. A defender is at . Let be the distance and the - angle. Given , as the ball leaves ,
the defender at starts along a straight line
at speed to intercept it. Ignore field boundaries.
Answer the following in terms of the given
parameters.

. Find the necessary condition for interception by the defender.

. An attacker waits on , distance from . Find when the ball is not intercepted by the defender initially at .

. An attacker is on , distance from . As the ball leaves , he runs at constant speed to receive it. Find when the defender initially at does not intercept it.

2.

Ground observations give satellite motion, then the motion of spacecraft or ground objects. Both require time and spatial coordinates. To simplify analysis and calculation, ignore Earth's spin and orbit; use Earth as an inertial frame.

. To measure satellite motion, ignore relativity. It has a radio transmitter and a precise atomic clock. Each signal contains its transmission time.

Several ground stations use these data, their known positions and clocks to find its position at each instant, hence its motion. For this system,